

Problem-Solution Fit

Date	26 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

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26 June 2026		OptiCrop: Smart Agricultural Production Optimization Engine	5 Marks					
1. CUSTOMER SEGMENT(S) [CS]  - Farmers - Agricultural Researchers - Agricultural Officers - Government Agriculture Departments - Agricultural Universities & Institutions	2. PROBLEMS / PAINS + ITS FREQUENCY [PR]  - Farmers often choose crops without scientific analysis. - Low crop yield due to unsuitable soil conditions. - Poor utilization of soil nutrients and water resources. - Crop failures caused by unfavorable environmental conditions. - Lack of easy access to intelligent crop recommendation systems.	3. TRIGGERS TO ACT [TR]  - Increasing demand for sustainable agriculture. - Need to improve crop productivity. - Requirement to reduce farming risks. - Growing adoption of AI in agriculture. - Need for data-driven farming decisions.	4. EMOTIONS BEFORE / AFTER [EM]  BEFORE - Confused about crop selection. - Worried about crop failure. - Fear of financial loss. - Uncertain about soil suitability. AFTER - Confident in crop selection. - Improved productivity. - Better resource utilization. - Increased farmer satisfaction.	5. AVAILABLE SOLUTIONS (Pros & Cons) [AS]  Traditional Farming Knowledge ✓ Experience-based ✗ May not always be accurate Agricultural Expert Consultation ✓ Reliable guidance ✗ Time-consuming and expensive Machine Learning-Based Crop Recommendation ✓ Fast, accurate, scalable, and data-driven ✓ Considers multiple environmental parameters				
6. CUSTOMER LIMITATIONS [CL]  - Limited knowledge of soil nutrient analysis. - Unpredictable weather conditions. - Limited access to agricultural experts. - Lack of technical knowledge. - Limited availability of real-time decision support.	7. BEHAVIOR + ITS INTENSITY [BE]  - Farmers want quick and accurate crop recommendations. - Agricultural researchers analyze crop-environment relationships. - Farmers increasingly use digital farming tools. - Demand for smart agriculture solutions is growing.	8. CHANNELS OF BEHAVIOR [CH]  <table border="1"> <thead> <tr> <th>ONLINE</th> <th>OFFLINE</th> </tr> </thead> <tbody> <tr> <td> - OptiCrop Web Application - Agricultural Portals - Government Agriculture Websites - Mobile Applications </td> <td> - Farmers - Agricultural Extension Officers - Krishi Vigyan Kendras (KVKs) - Agricultural Research Centers </td> </tr> </tbody> </table>		ONLINE	OFFLINE	- OptiCrop Web Application - Agricultural Portals - Government Agriculture Websites - Mobile Applications	- Farmers - Agricultural Extension Officers - Krishi Vigyan Kendras (KVKs) - Agricultural Research Centers	9. PROBLEM ROOT / CAUSE [RC]  - Lack of scientific crop selection. - Soil nutrient imbalance. - Climate variability. - Manual decision-making. - Limited access to intelligent recommendation systems.
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10. YOUR SOLUTION [SL]  Develop an OptiCrop: Smart Agricultural Production Optimization Engine using Machine Learning. The system analyzes Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall to recommend the most suitable crop for cultivation. The solution improves crop productivity, optimizes resource utilization, supports sustainable farming, and assists farmers in making accurate, data-driven agricultural decisions through a Flask-based web application.								